



CIVITAS indicators

Share of deliveries made by Battery-Electric and Hydrogen Fuel-Cell vans (ENV_FR_EL)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
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TOPIC

Freight

IMPACT

Zero-emission deliveries

Improved environmental performance of urban freight distribution through an increase in the use of battery-electric and hydrogen fuel-cell vans for last-mile deliveries.

ENV_FR

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Urban freight measures frequently aim to reduce the environmental impact of delivery operations while maintaining the ability of logistics operators to provide reliable and efficient services. Measures such as low-emission zones, zero-emission zones, charging infrastructure, fleet renewal schemes, urban consolidation centres and other sustainable logistics solutions can encourage operators to replace conventionally fuelled delivery vans with battery-electric and hydrogen fuel-cell vehicles. Increased adoption of these vehicles can help reduce greenhouse gas emissions, improve urban air quality and reduce noise impacts associated with urban freight activity.

This indicator measures the share of completed deliveries performed by battery-electric (BEV) and hydrogen fuel-cell (FCEV) vans within a defined geographical area and reference period. **It is a relevant indicator when the policy action aims to increase the uptake of zero-emission delivery vans in urban freight operations. A successful action is reflected in a HIGHER value of the indicator after the experiment compared to the BAU case.** Deliveries performed by cargo bikes and other zero-emission modes are excluded from this indicator and assessed through separate indicators where relevant.

DESCRIPTION

The indicator assesses the extent to which urban freight deliveries are performed using battery-electric and hydrogen fuel-cell vans rather than internal combustion engine delivery vehicles. It reflects the adoption of zero-emission van technologies by logistics operators and captures the combined effects of policy measures, operational changes and investment decisions that support cleaner urban freight transport.

The indicator is calculated by dividing the total number of deliveries completed by battery-electric and hydrogen fuel-cell vans during the observation period by the total number of deliveries completed during the same period. Deliveries carried out by cargo bikes, walking couriers or other alternative zero-emission modes are not included in the numerator.

The indicator is expressed as **% of deliveries**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously**. The user computes the indicator value outside the CIVITAS Evaluation tool according to the method described below and enters the final result into the tool.

Method 1		
Survey delivery companies	Significance: 0.75	
INPUTS		
The following information is needed to compute the indicator:		
Total number of deliveries completed by battery-electric vans during the chosen observation period;		

- Total number of **deliveries completed by hydrogen fuel-cell vans** during the observation period;
- Total number of **deliveries completed** during the observation period;

EQUATIONS

The equation that should be applied to quantify the indicator is:

$$ZED_{vans} = \frac{BEVDeliveries + FCEVDeliveries}{Deliveries} \times 100$$

Where:

ZED_{vans} = Share of deliveries made by battery-electric and hydrogen fuel-cell vans (%)

BEVDeliveries = Number of deliveries completed by battery-electric vans during the observation period

FCEVDeliveries = Number of deliveries completed by hydrogen fuel-cell vans during the observation period

Deliveries = Total number of completed deliveries during the observation period

ALTERNATIVE INDICATORS

This indicator measures the uptake of zero-emission van technologies for urban freight deliveries. To assess the environmental performance of urban freight operations in terms of carbon emissions, the framework also provides **ENV_FR_CO1**: CO₂ emissions per km and **ENV_FR_CO2**: CO₂ emissions per delivery. To assess the use of non-motorised delivery vehicles, the framework provides **TRA_FR_MA2**: Share of deliveries made by non-motorised vehicles and **TRA_FR_IL1**: Share of deliveries by cargo bike.